

RESEARCH FOCUS

- Theoretical foundations and applications of machine learning for discovering physical laws from observation data in complex systems.
- Stochastic multiscale modeling and uncertainty quantification of granular materials, fluid dynamics, and multi-agent systems.
- Music technology: algorithmic composition, intelligent musical interface design and development.

EDUCATION

Stanford University Ph.D., Granular Materials, Advisor: Daniel Tartakovsky Thesis: <i>Stochastic multiscale modeling of complex materials</i> Department of Energy Science and Engineering	Palo Alto, CA 2017–2020
University of California San Diego M.S./Ph.D. in Fluid Mechanics Department of Mechanical and Aerospace Engineering	San Diego, CA 2013–2016
American University of Beirut B.Eng. in Mechanical Engineering	Beirut, Lebanon 2009–2013
Lebanese National Higher Conservatory of Music Bacc. in Piano Performance	Beirut, Lebanon 2003–2013

EXPERIENCE

American University of Beirut Assistant Professor	Beirut, Lebanon 2024–Current
– Leading the Music Intelligence Lab and Data-Driven Modeling Lab at the Department of Mechanical Engineering, MSFEA. – Co-founding faculty member of the School of Computing and Data Sciences, previously AI Hub. – Center for Advanced Mathematical Sciences Fellow	
University of Washington Postdoctoral Fellow, AI Institute in Dynamic Systems, with S. Brunton & J. N. Kutz	Seattle, WA 2020–2023
– Data-driven discovery of nonlinear dynamics in complex systems	
American University of Beirut Visiting Lecturer, Center For Advanced Mathematical Sciences	Beirut, Lebanon Winter 2022
– Instructor of graduate course in data-driven methods in science and engineering – Research on data-driven discovery of contaminant transport in mediterranean	

Los Alamos National Lab
Earth and Environmental Sciences

Los Alamos, NM
Summers 2014-16

- Computational agent based modeling for socio-hydrological simulations

University of California San Diego
Music Technology Lab

San Diego, CA
2015-2016

- Design, building and machine learning of a glove musical instrument

University of California Berkeley
Swarm Lab

Berkeley, CA
Summer 2012

- Finless micro-rocket: Modeling, control, manufacturing of micro-rockets without fins

American University of Beirut
Aerosols Research Lab, with Alan L. Shihadeh

Beirut, Lebanon
Summer 2013

- Experimental data collection & analysis of water-pipe hookah toxicity studies

American University of Beirut
Microflows and Microscale Heat Transfer Lab, with Issam Lakkis

Beirut, Lebanon
2011-2013

- Modeling and simulation of a lab-on-a-chip microfluidic transistor
- Vortex methods for slip boundary conditions

PUBLICATIONS

1. Sadek, N. & **Bakarji, J.**, “The Circle of Fifths as Latent Geometry in Bach’s Well-Tempered Clavier”, *1st International Workshop on Emerging AI Technologies for Music, PMLR*, (2026)
2. Germany, J., Abdo, E. & **Bakarji, J.**, “EML Trees Are Universal Approximators”, *preprint arXiv:2606.23179*, (2026)
3. Germany, J., **Bakarji, J.** & Najem, S., “Discovery of Symbolic Hamiltonian Expressions with Buckingham-Symplectic Networks”, *preprint arXiv:2604.00576*, (2026)
4. Riva, S., Introini, C., Cammi, A., Price, D., Yermakov, A., Zhao, Y., Wyder, P.M., Goldfeder, J., Williams, J., Rude, A.S., Tomasetto, M., Germany, J., **Bakarji, J.**, Maierhofer, G., Cranmer, M. & Kutz, J.N., “CTF4Nuclear: Common Task Framework for Nuclear Fission and Fusion Models”, *preprint arXiv:2605.15549*, (2026)
5. Wyder, P.M., Goldfeder, J., Yermakov, A., Zhao, Y., Riva, S., Williams, J., Zoro, D., Rude, A.S., Tomasetto, M., Germany, J., **Bakarji, J.**, Maierhofer, G., Cranmer, M. & Kutz, J.N., “Common Task Framework for a Critical Evaluation of Scientific Machine Learning Algorithms”, *Advances in Neural Information Processing Systems (NeurIPS)*, (2025)
6. Yermakov, A., Zhao, Y., Denolle, M., Ni, Y., Wyder, P.M., Goldfeder, J., Riva, S., Williams, J., Zoro, D., Rude, A.S., Tomasetto, M., Germany, J., **Bakarji, J.**, Maierhofer, G., Cranmer, M. & Kutz, J.N., “The Seismic Wavefield Common Task Framework”, *preprint arXiv:2512.19927*, (2025)
7. Haschka, T. & **Bakarji, J.**, “Semantic Tree Inference on Text Corpa using a Nested Density Approach together with Large Language Model Embeddings”, *preprint arXiv:2512.23471, submitted to Data Mining and Knowledge Discovery*, (2025)

8. Klishin, A.A., **Bakarji, J.**, Kutz, J.N. & Manohar, K., “Statistical Mechanics of Dynamical System Identification”, *Physical Review Research*, (2025)
9. Chandra, A., **Bakarji, J.** & Tartakovsky, D.M., “Role of physics in physics-informed machine learning”, *Journal of Machine Learning for Modeling and Computing*, (2024)
10. **Bakarji, J.**, “Distilling data into code”, *Nature Computational Science*, (2024)
11. **Bakarji, J.**, Champion, K., Kutz, J.N. & Brunton, S.L., “Discovering governing equations from partial measurements with deep delay autoencoders”, *Proceedings of the Royal Society A*, (2023)
12. **Bakarji, J.**, Callaham, J., Brunton, S.L. & Kutz, J.N., “Dimensionally consistent learning with Buckingham Pi”, *Nature Computational Science*, (2022)
13. **Bakarji, J.** & Tartakovsky, D.M., “Data-driven discovery of coarse-grained equations”, *Journal of Computational Physics*, 434, 110219, (2021)
14. **Bakarji, J.** & Tartakovsky, D.M., “Stochastic pore collapse models in granular materials”, *preprint arXiv:2002.12479*, (2020)
15. **Bakarji, J.**, “Machine Learning for a Music Glove Instrument”, *preprint arXiv:2001.09551*, (2020)
16. **Bakarji, J.** & Tartakovsky, D.M., “Microstructural heterogeneity drives reaction initiation in granular materials”, *Applied Physics Letters*, 114 (25), 254101, (2019)
17. **Bakarji, J.**, O'Malley, D. & Vesselinov, V.V., “Agent-based socio-hydrological hybrid modeling for water resource management”, *Water Resources Management*, 31 (12), 3881-3898, (2017)
18. **Bakarji, J.** & Tartakovsky, D.M., “On the use of reverse Brownian motion to accelerate hybrid simulations”, *Journal of Computational Physics*, 334, 68-80, (2017)
19. **Bakarji, J.**, Keniar, K., Cheikh, M. & Lakkis, I., “A reduced-order model of a microfluidic transistor”, *ASME International Mechanical Engineering Congress and Exposition*, (Vol. 56390, p. V010T11A025), (2013)

TEACHING

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| <ul style="list-style-type: none"> ● Online Lectures - YouTube
https://www.youtube.com/c/JosephBakarji | 2020-Present |
| <ul style="list-style-type: none"> ● Instructor at the American University of Beirut
<i>Dynamics</i> | 2025-Present |
| <ul style="list-style-type: none"> ● Instructor at the American University of Beirut
<i>Data-Driven Modeling in Science and Engineering</i> | 2024-Present |
| <ul style="list-style-type: none"> ● Instructor at the American University of Beirut
<i>Introduction to Machine Learning</i> | 2024-Present |
| <ul style="list-style-type: none"> ● Instructor at the University of Washington
<i>Deep Learning Tutorial, at AI Institute CTF Workshop</i> | Winter 2023 |
| <ul style="list-style-type: none"> ● Instructor at the American University of Beirut
<i>Data-driven Methods in Science and Engineering</i> | Winter 2022 |

• Research Mentorship at University of Washington <i>Joseph Williams (PhD), Munzir Hashim (MS), Gabriel Bufolo (PhD)</i>	2020-Present
• Instructor at GTLogic <i>Introduction to Machine Learning</i>	Summer 2021
• Instructor at University of Washington <i>Deep Learning for Fluid Dynamics</i>	Spring 2021
• Instructor at the American University of Beirut <i>Introduction to Multiphase Flow in Porous Media</i>	Summer 2018
• Teaching Assistant at Stanford University <i>Applied Mathematics for Reservoir Engineering</i>	Winter 2017
• Teaching Assistant at UC San Diego <i>Introduction to Uncertainty Quantification</i>	Fall 2017
• Teaching Assistant at UC San Diego <i>Introduction to Dynamics</i>	Winter 2016
• Teaching Assistant at UC San Diego <i>Dynamics and Control of Aircraft Vehicles</i>	Winter 2015
• Teaching Assistant at UC San Diego <i>Aircraft Engineering Design</i>	Spring 2014
• Teaching Assistant at UC San Diego <i>Introduction to Music Technology</i>	Summer 2015
• Instructor at Lebanese National Higher Conservatory of Music <i>Piano performance</i>	2011-2013

AWARDS

• Center for Advanced Mathematical Sciences Fellowship, American University of Beirut	2022
• Henry J. Ramey, Jr. Fellowship Award, Stanford University	2020
• Frank G. Miller Fellowship Award, Stanford University	2018
• MATLAB Award, Big Earth Hackathon, Stanford University	2017
• Best Poster Award, Los Alamos National Laboratory	2016
• SME Boat Competition (1st place), American University of Beirut	2013

SPEAKER

• AAI, speaker (Singapore)	2026
• NeurIPS, poster (San Diego, USA)	2025
• ENUMATH, Invited speaker (Heidelberg, Germany)	2025
• UNCECOMP, speaker and co-chair (Rhodes, Greece)	2025
• Build with AI, Invited speaker at Google Developer Group (Beirut, Lebanon)	2025
• AICOMAS, Invited Keynote speaker (Paris, France)	2025

• AUB ME and EECE Seminar, speaker (Beirut, Lebanon)	2024
• ECCOMAS, Invited speaker (Lisbon, Portugal)	2024
• IDEAS Conference, AUB (Beirut, Lebanon)	2024
• CAMS colloquium, AUB (Beirut, Lebanon)	2024
• Data-driven modeling seminar, AUB (Beirut, Lebanon)	2024
• SIAM WCCM, Mini-symposium co-organizer and speaker (Vancouver, Canada)	2024
• SIAM Mathematics of CSE, invited speaker (Amsterdam, NL)	2023
• University of Brasilia Mechanics Seminar, invited speaker (Brasilia, Brazil)	2022
• Columbia University, LEAP ML Climate Journal Club, invited speaker (Virtual, Manhattan, NY)	2022
• Applied and Computational Mathematics Seminar, invited speaker (Tulane, LA)	2022
• AUB mechanical engineering seminar, invited talk (Beirut, Lebanon)	2022
• CWI Amsterdam seminar, invited speaker (Virtual)	2022
• SIAM Mathematics of Data Science, invited speaker (San Diego, CA)	2022
• MPML University of Lisbon, invited speaker (Lisbon, Portugal)	2022
• US National Congress for Ap. and Theoretical Mechanics, invited speaker (Austin, TX)	2022
• Oak Ridge National Laboratory workshop on AI for robust engineering (Oak Ridge, TN)	2022
• University of Southern California, invited speaker (Los Angeles, CA)	2022
• APS Division of Fluid Dynamics (Phoenix, AZ)	2021
• SIAM Annual Meeting (Virtual)	2021
• Carnegie Mellon University, invited Speaker Scientific Machine Learning (Virtual)	2021
• SIAM Computational Science and Engineering Conference (Virtual)	2021
• SIAM Computational Science and Engineering Conference (Spokane, WA)	2019
• Multiscale Modeling Workshop AFOSR (Eglin, FL)	2018
• Air Force Research Lab Annual Review (Washington D.C.)	2016
• SIAM Geosciences Conference (Stanford, CA)	2014

SERVICES

• Member of the founding committee for the School of Computing and Data Sciences (SCDS)	Spring 2024-Present
• Mechanical Engineering Undergraduate Curriculum Committee for Mathematics	Spring 2024-Present
• Mechanical Engineering Graduate Admissions Committee	Spring 2024-Present
• MSFEA Graduate Admissions Committee	Spring 2024-Present
• Faculty Search Committee - Mechanical Engineering Dept.	2024
• Graduate Admissions Committee - Mechanical Engineering Dept.	Spring 2024
• MSFEA/CAMS host and organizer of international speakers: Sherif Maktabi (Amazon), Mark Saroufim (Meta), Nathan Kutz (University of Washington), Ryan Raut (Allen Institute)	Spring 2024-Present
• AUB Bioengineering Hackathon jury and supervisor	Spring 2024

- AUB Mathematics High School Summer Camp - Introduction to Machine Learning Tutorial Summer 2024
- Mechanical Engineering Program Preparation, AUB Mediterraneo Fall-Summer 2024
- MSFEA Experiential Learning Retreat Fall 2024
- CAMS Data-driven Modeling for the Sciences seminar co-organizer 2024-Present
- AI Institute Data-driven Methods in Science and Engineering seminar co-organizer 2020-2023
- Membership in: American Physical Society (APS), Society of Industrial and Applied Mathematics (SIAM).
- Graduate Committee Member: Omar El Sagheer (M.S. GPCS), Jennyfer Karam (Ph.D. ME) Spring 2024-Present
- Reviewer at: Nature Communications, Proceedings of the Royal Society A, NeurIPS, Nature Computational Science, Nature Communications Physics, Neural Networks Elsevier, Digital Discovery, Scientific Reports, European Conference for Artificial Intelligence, IEEE Transactions on Cybernetics, American Meteorological Society, IEEE/RSJ International Conference on Intelligent Robots and Systems, University Research Board (AUB), Journal of the Atmospheric Sciences, Journal of Geophysical Research - Machine Learning and Computation, AGU, Journal of Machine Learning for Modeling and Computing, Journal of Computational Physics, AIP Physics of Fluids, Journal of Fluid Mechanics, Physica D: Nonlinear Phenomena, Computer Methods in Applied Mechanics and Engineering.
- The Scientist Pen-Pal program for high school students, Stanford 2018

RESEARCH SUPERVISION

- Khyam Allami (Postdoc) Fall 2024 - Present
- Ali Yehya (MechE Ph.D. w/ Dr. Issam Lakkis) Spring 2024 - Present
- Fatima Fneich (Ph.D. w/ Dr. Arij Daou) Fall 2024 - Present
- Marwa Al Korhanie (GPCS M.S. w/ Dr. Arij Daou) Fall 2024 - Present
- Lama Sleem (GPCS M.S. w/ Dr. Arij Daou) Fall 2024 - Present
- Lana Wayzani (GPCS M.S.) Fall 2024 - Present
- Roni Trad (GPCS M.S.) Spring 2024 - Present
- Issar Amro (GPCS M.S. w/ Dr. Sara Najem) Fall 2024 - Present
- Oussama Ibrahim (GPCS M.S. w/ Dr. Sara Najem) Spring 2024 - Present
- Rebal Abou Fadel (Undergraduate) Fall 2025
- Joe Germany (Undergraduate) Spring 2024 - Present
- Najla Sadek (Undergraduate) Spring 2025 - Present
- Saja Borgol (Undergraduate) Spring 2025
- Ibrahim El Khansa (Undergraduate) Spring 2025
- Artur Baboudjian (Undergraduate) Spring 2025
- Mohamad Lakkis (Undergraduate) Spring 2025
- Mohamad Bazzi (Undergraduate) Spring 2025
- Zeinab Takash (Undergraduate) Spring 2025
- Mohamad Khaled Charaf (Undergraduate) Fall 2024
- Amir Soufi (M.S.) Fall 2024

SOFTWARE

Available on www.github.com/josephbakarji/. Selected Repositories:

- **deep-delay-autoencoder** : Discovers high dimensional models from 1D data using deep delay autoencoders
- **bucki-data**: Machine learning algorithms for discovering dimensionless groups from simulation and experimental data
- **granular-compaction**: Monte Carlo 1D dynamic compaction of a two-phase multiscale granular material
- **NewsAndTime**: Machine learning algorithm for predicting date of publication from content of news article
- **learning-pdf**: Sparse identification of PDF equation and closure models based on Monte Carlo simulations
- **Socio-hydro-abm**: Agent based simulations and control of socio-hydrological models
- **Gluvn**: Sensor I/O and machine learning for Music glove instrument

LANGUAGES

- **Programming:** Python, Matlab, Julia, C/C++, Fortran, Labview, Assembly.
- **Spoken:** English (Fluent), French (Fluent), Arabic (Fluent), Italian (Intermediate).

MUSIC

Public composition channel: www.soundcloud.com/user-343358875

- International composition conference for young composers, Cergy Pontoise, France 2011
- Composition quartet performance by Quatuor de Beyrouth: Life quartet 2006
- Composition percussion performance by Symblema: The lightness of chaos 2012
- Second prize, Piano improv, AUB's got talent 2012
- Third prize, Babikian piano competition 2009
- First prize, Lions Club piano competition 2008
- Second prize, LaMajeur piano competition 2008
- Piano concert, Lebanese National Conservatory 2008
- Master classes with S. Bunin, E. Sombart, W. Howrani and W. Mousallem